

A positive Hilbert–Schmidt cone counterexample to a common quasi-interior lower bound

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Source Question

The source paper is J. Glueck and A. Mironchenko, *Stability criteria for positive semigroups on ordered Banach spaces*, arXiv:2210.13566. After Theorem 8.4, in the setting of an ordered Hilbert space (H, H^+) satisfying

$$H^+ \subseteq (H')^+,$$

the authors note that their hypothesis gives quasi-interior points u, v whenever there is a quasi-interior point w with $u, v \geq w$. They then write that they do not know whether the converse holds: given arbitrary quasi-interior points $u, v \in H^+$, must there always be a quasi-interior point $w \in H^+$ such that $u, v \geq w$?

Result

The answer is negative. In fact, the following example gives two quasi-interior points with no nonzero common positive lower bound.

Theorem 1. *There is an ordered real Hilbert space $(\mathcal{H}_{\text{sa}}, \mathcal{H}_{\text{sa}}^+)$ with self-dual cone and two quasi-interior points $A, B \in \mathcal{H}_{\text{sa}}^+$ such that*

$$0 \leq C \leq A, B \implies C = 0.$$

Consequently there is no quasi-interior point C satisfying $A, B \geq C$.

Proof Intuition

The positive cone of self-adjoint Hilbert–Schmidt operators behaves like the infinite-dimensional positive semidefinite matrix cone. A positive Hilbert–Schmidt operator is quasi-interior precisely when it has no kernel: its square-root range is then dense and supplies enough rank-one positive operators to densely generate the Hilbert–Schmidt space.

The obstruction is that two dense operator ranges can be transverse. We choose an injective diagonal compact operator whose square-root range is

$$R = \{x = (x_n) \in \ell_2 : \sum n^2 |x_n|^2 < \infty\}.$$

A Baire argument gives a unitary U such that $R \cap UR = \{0\}$. If A has square-root range R and $B = UAU^*$ has square-root range UR , Douglas' range inclusion lemma forces every positive common lower bound C to have

$$\text{Ran}(C^{1/2}) \subseteq R \cap UR = \{0\}.$$

Thus $C = 0$, even though both A and B are quasi-interior.

A transverse weighted range

We first record the elementary Hilbert-space ingredient used in the construction.

Lemma 1 (A transverse compactly embedded dense range). *Let $E = \ell_2(\mathbb{N})$ and*

$$R = \left\{ x = (x_n) \in E : \sum_{n=1}^{\infty} n^2 |x_n|^2 < \infty \right\}.$$

There is a unitary operator U on E such that

$$R \cap UR = \{0\}.$$

Proof. Endow R with the Hilbert norm

$$\|x\|_R^2 = \sum_{n=1}^{\infty} n^2 |x_n|^2.$$

The inclusion $R \hookrightarrow E$ is compact. For $m \in \mathbb{N}$ set

$$K_m = \{x \in R : \|x\|_E = 1, \|x\|_R \leq m\}.$$

Each K_m is compact in E , and the unit sphere of R is $\bigcup_m K_m$.

Let $\mathcal{U}(E)$ be the unitary group with the operator-norm topology. For $m, n \in \mathbb{N}$ define

$$\mathcal{E}_{m,n} = \{V \in \mathcal{U}(E) : VK_m \cap K_n \neq \emptyset\}.$$

The sets $\mathcal{E}_{m,n}$ are closed. Indeed, if $V_j \rightarrow V$ in norm and $V_j x_j \in K_n$ for $x_j \in K_m$, compactness of K_m gives a subsequence $x_j \rightarrow x \in K_m$, hence $V_j x_j \rightarrow Vx \in K_n$.

We show that each $\mathcal{E}_{m,n}$ has empty interior. It suffices to prove the following small perturbation fact: if K, L are compact subsets of the unit sphere of an infinite-dimensional Hilbert space, V_0 is unitary, and $\varepsilon > 0$, then there is a unitary V with $\|V - V_0\| < \varepsilon$ and $VK \cap L = \emptyset$.

Put $K_0 = V_0 K$. Choose $0 < \theta < \varepsilon/2$. Since $K_0 \cup L$ is compact, there is a finite-dimensional subspace $F \subset E$ such that every point of $K_0 \cup L$ is within δ of F , where $\delta > 0$ will be chosen so small that

$$\sin \theta (1 - \delta) > 2\delta.$$

Choose an isometric copy $J : F \rightarrow F^\perp$. Let W be the unitary which rotates F into JF through angle θ ,

$$Wf = \cos \theta f + \sin \theta Jf \quad (f \in F),$$

with the inverse rotation on JF and the identity on $(F \oplus JF)^\perp$. Then $\|W - I\| < \varepsilon$. If $x \in K_0$ and $y \in L$, write P for the orthogonal projection onto F . The JF -component of Wx has norm at

least $\sin \theta \|Px\| - \|(I - P)x\|$, which is $> \delta$ by the choice of δ , while the JF -component of y has norm at most δ . Thus $Wx \neq y$. Hence $WK_0 \cap L = \emptyset$, and $V = WV_0$ has the desired properties.

Applying this perturbation fact with $K = K_m$ and $L = K_n$ proves that each $\mathcal{E}_{m,n}$ is nowhere dense. Since $\mathcal{U}(E)$ is a complete metric space in operator norm, Baire's theorem gives a unitary

$$U \notin \bigcup_{m,n=1}^{\infty} \mathcal{E}_{m,n}.$$

If $0 \neq x \in R \cap UR$, then after normalizing x to norm one, $x \in K_n$ for some n and $x = Uy$ for a norm-one $y \in R$, with $y \in K_m$ for some m . This would put U in $\mathcal{E}_{m,n}$, a contradiction. $\square$

The Hilbert–Schmidt cone

Let $E = \ell_2(\mathbb{N})$, now viewed as a real Hilbert space. Let $\mathcal{H}_{\text{sa}}$ be the real Hilbert space of all self-adjoint Hilbert–Schmidt operators on E , with inner product

$$\langle S, T \rangle_{\mathcal{H}_{\text{sa}}} = \text{Tr}(ST).$$

Let $\mathcal{H}_{\text{sa}}^+$ be the cone of positive semidefinite Hilbert–Schmidt operators.

Lemma 2. *The cone $\mathcal{H}_{\text{sa}}^+$ is self-dual in $\mathcal{H}_{\text{sa}}$. Moreover, for $A \in \mathcal{H}_{\text{sa}}^+$, the following are equivalent:*

$$A \text{ is a quasi-interior point of } \mathcal{H}_{\text{sa}}^+ \iff \ker A = \{0\}.$$

Proof. Self-duality follows from rank-one tests. If $A \in \mathcal{H}_{\text{sa}}$ satisfies $\text{Tr}(AB) \geq 0$ for every $B \in \mathcal{H}_{\text{sa}}^+$, then testing against $B = x \otimes x$ gives $\langle Ax, x \rangle \geq 0$ for all $x \in E$, so $A \geq 0$. The converse implication $\text{Tr}(AB) \geq 0$ for $A, B \geq 0$ follows by writing the trace as the trace of $A^{1/2}BA^{1/2}$, first for finite-rank approximants and then by approximation.

Assume first that $\ker A \neq \{0\}$ and choose $0 \neq h \in \ker A$. If $S \in [-cA, cA]$ for some $c > 0$, then $cA + S \geq 0$ and $cA - S \geq 0$. Since both positive operators have quadratic form value 0 at h , they both annihilate h ; hence $Sh = 0$. Thus the order ideal $\bigcup_{c>0} [-cA, cA]$ is contained in the closed proper subspace of Hilbert–Schmidt operators that annihilate h , and is not dense. So A is not quasi-interior.

Conversely, assume $\ker A = \{0\}$. Then $\text{Ran}(A^{1/2})$ is dense in E . For $z = A^{1/2}y$ one has the rank-one domination

$$z \otimes z \leq \|y\|^2 A$$

as quadratic forms, since

$$|\langle x, z \rangle|^2 = |\langle A^{1/2}x, y \rangle|^2 \leq \|y\|^2 \langle Ax, x \rangle.$$

Hence every rank-one projection $z \otimes z$ with $z \in \text{Ran}(A^{1/2})$ lies in the positive part of the principal ideal generated by A . By polarization, the real linear span of these rank-one projections contains the finite-rank self-adjoint operators generated by vectors in the dense subspace $\text{Ran}(A^{1/2})$. Those finite-rank self-adjoint operators are dense in $\mathcal{H}_{\text{sa}}$. Therefore the principal ideal generated by A is dense, so A is quasi-interior. $\square$

Counterexample

Let D be the positive diagonal operator on $E = \ell_2(\mathbb{N})$ given by

$$De_n = \frac{1}{n^2}e_n.$$

Then $D \in \mathcal{H}_{\text{sa}}^+$, D is injective, and

$$\text{Ran}(D^{1/2}) = \left\{ x = (x_n) \in \ell_2 : \sum_{n=1}^{\infty} n^2 |x_n|^2 < \infty \right\} = R.$$

Choose, by Lemma 1, a unitary U on E such that $R \cap UR = \{0\}$, and define

$$A = D, \quad B = UDU^*.$$

Both A and B are injective positive Hilbert–Schmidt operators, hence both are quasi-interior points of $\mathcal{H}_{\text{sa}}^+$ by Lemma 2.

Now suppose $0 \leq C \leq A$ and $0 \leq C \leq B$ for some $C \in \mathcal{H}_{\text{sa}}^+$. Douglas’ range inclusion lemma says that for bounded operators S, T on a Hilbert space, $\text{Ran } S \subseteq \text{Ran } T$ is equivalent to $SS^* \leq \lambda TT^*$ for some $\lambda \geq 0$. Applying this with $S = C^{1/2}$ and $T = A^{1/2}$, and then with $T = B^{1/2}$, yields

$$\text{Ran}(C^{1/2}) \subseteq \text{Ran}(A^{1/2}) = R, \quad \text{Ran}(C^{1/2}) \subseteq \text{Ran}(B^{1/2}) = UR.$$

Thus

$$\text{Ran}(C^{1/2}) \subseteq R \cap UR = \{0\},$$

so $C^{1/2} = 0$ and $C = 0$.

This proves that the quasi-interior points A and B have no nonzero common positive lower bound. In particular, they do not dominate a common quasi-interior point.

Verification and Limitations

The cone $\mathcal{H}_{\text{sa}}^+$ is self-dual, so the source hypothesis $H^+ \subseteq (H')^+$ is satisfied with equality. The example is not a Banach lattice cone and has empty interior, which is precisely where the L^2 lattice infimum argument from the source fails.

The counterexample does not address the separate open problem, cited in the source from Glueck–Weber, asking whether quasi-interior and almost-interior points coincide for every generating normal cone. It only answers the common-lower-bound question following Theorem 8.4.

Novelty Check

Before promotion, the run indexes were searched for arXiv:2210.13566, quasi-interior points, ordered Hilbert spaces, positive Hilbert–Schmidt cones, self-dual cones, operator ranges, and Douglas range inclusion. No exact prior packet or duplicate was found. Web searches for close phrases such as “quasi-interior common lower bound ordered Hilbert space”, “positive Hilbert-Schmidt quasi-interior”, and “operator ranges trivial intersection Hilbert space” found related background but no explicit answer to this source question.

References

1. J. Glueck and A. Mironchenko, *Stability criteria for positive semigroups on ordered Banach spaces*, arXiv:2210.13566.
2. R. G. Douglas, *On majorization, factorization, and range inclusion of operators on Hilbert space*, Proceedings of the American Mathematical Society 17 (1966), 413–415.